

TOHMAJARVI KEMIE, Finland

WMO: 028320

Lat: 62.235N

Lon: 30.347E

Elev: 90

StdP: 100.25

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.4	-25.8	-32.8	0.2	-29.3	-28.7	0.3	-25.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	27.2	19.1	25.1	17.9	23.2	16.8	20.6	25.2	19.2	23.5	18.0	21.7	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	13.9	23.1	17.6	12.7	21.4	16.3	11.7	19.9	59.9	25.2	55.1	23.7	51.1	21.8	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	-31.0	28.8	4.3	3.0	-34.1	30.9	-36.6	32.7	-39.1	34.4	-42.2	36.6
			-31.5	22.0	4.2	2.2	-34.4	23.6	-36.9	24.9	-39.2	26.1	-42.2	27.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.4	-10.3	-7.9	-4.2	2.4	9.2	13.1	16.4	14.4	9.7	3.1	-0.8	-4.7
	DBStd	10.07	8.27	7.25	5.36	3.69	4.30	3.72	3.40	3.19	3.37	3.90	4.46	6.53
	HDD10.0	2924	631	500	439	230	66	13	2	4	47	215	324	454
	HDD18.3	5469	889	733	697	479	283	162	79	128	260	472	574	713
	CDD10.0	524	0	0	0	1	42	106	199	139	37	1	0	0
	CDD18.3	28	0	0	0	0	1	4	19	5	0	0	0	0
	CDH23.3	402	0	0	0	0	37	74	224	67	0	0	0	0
	CDH26.7	74	0	0	0	0	3	12	48	11	0	0	0	0
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(15) Precipitation	PrecAvg	672	49	40	35	31	45	70	73	81	62	67	60	59
	PrecMax	900	101	85	67	71	110	131	160	155	177	148	109	126
	PrecMin	537	12	4	5	6	17	34	12	26	24	16	9	23
	PrecStd	89	21	20	15	16	22	28	37	32	37	31	24	23
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.9	4.0	9.0	17.6	26.9	28.3	30.5	28.3	20.6	12.5	9.1	5.8
		MCWB	2.3	2.3	3.7	9.6	15.4	18.1	21.4	20.5	16.1	10.9	8.3	5.3
	2%	DB	1.9	2.4	5.5	13.7	24.0	25.5	27.9	25.1	18.2	11.0	6.9	3.7
		MCWB	1.4	1.3	2.1	7.7	14.4	17.7	20.2	18.6	14.4	9.8	6.2	3.2
	5%	DB	0.7	1.4	3.7	11.3	21.1	23.1	25.9	22.8	16.5	9.9	5.9	2.4
		MCWB	0.2	0.7	1.3	6.0	13.4	16.0	19.2	17.3	13.7	8.6	5.5	1.8
	10%	DB	0.1	0.5	2.3	9.2	18.3	21.0	23.8	20.8	15.2	8.8	4.6	1.4
		MCWB	-0.3	0.1	0.6	4.9	11.3	14.9	18.0	16.4	13.2	7.7	4.2	1.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.5	2.6	4.2	10.8	17.3	21.3	22.9	21.6	16.6	11.5	8.4	5.5
		MCDB	2.7	3.4	6.9	14.7	23.6	25.5	28.4	26.3	19.2	12.4	9.0	5.8
	2%	WB	1.5	1.6	2.7	8.6	15.4	18.9	21.2	19.2	15.5	10.2	6.4	3.2
		MCDB	1.8	2.2	4.7	13.4	21.3	23.7	26.1	23.3	17.3	10.8	6.8	3.6
	5%	WB	0.6	0.8	1.8	6.6	13.9	16.9	20.1	18.2	14.4	8.9	5.4	2.0
		MCDB	0.9	1.3	3.2	10.4	19.9	20.7	24.6	21.8	15.9	9.5	5.8	2.3
	10%	WB	-0.1	0.2	1.0	5.2	12.3	15.6	18.8	17.1	13.5	7.8	4.3	1.1
		MCDB	0.2	0.6	2.0	8.3	17.1	20.1	22.9	20.2	14.8	8.6	4.6	1.3
(35) Mean Daily Temperature Range	5% DB	MDBR	6.4	6.9	9.1	10.0	12.3	11.0	11.0	10.2	7.9	5.7	3.9	4.4
		MCDBR	3.2	4.9	9.5	14.4	17.6	15.3	14.8	14.1	10.6	7.2	4.8	3.6
		MCWBR	3.2	4.5	6.8	8.6	9.1	8.0	7.8	8.4	7.3	6.1	4.5	3.4
	5% WB	MCDBR	3.1	4.4	7.1	12.3	14.9	13.1	13.2	12.0	8.7	6.0	4.6	3.6
		MCWBR	3.1	4.2	5.4	7.7	8.3	7.6	7.6	7.4	6.5	5.5	4.4	3.5
(40) Clear-Sky Solar Irradiance	taub		0.241	0.290	0.316	0.355	0.355	0.349	0.376	0.359	0.337	0.317	0.259	0.216
	taud		2.021	2.101	2.166	2.296	2.438	2.483	2.421	2.451	2.503	2.398	2.117	1.974
	Ebn at Noon		416	645	780	829	865	877	838	822	774	637	393	253
	Edh at Noon		36	70	96	104	99	96	100	89	71	54	33	20
(44) All-Sky Solar Radiation	RadAvg		0.21	0.78	2.24	3.68	4.92	5.31	5.07	3.87	2.17	0.84	0.21	0.10
	RadStd		0.03	0.18	0.30	0.40	0.52	0.33	0.47	0.43	0.27	0.12	0.04	0.01

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (25 neighbors)	+0.54	N/A	N/A	N/A	N/A	N/A	-185	N/A	N/A	N/A

Nomenclature: See separate page